MINI PROJECT REPORT

A Mini Project Report on **Job Vacancy Registration Portal**
submitted to the
Department of Computer Science
St. Joseph's College (Autonomous)
Affiliated to Bharathidasan University, Tiruchirappalli

in partial fulfilment of the requirements for the award of the Degree of
BACHELOR OF SCIENCE IN COMPUTER SCIENCE

BY

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Department of Computer Science
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Accredited at A++ Grade (Cycle IV) by NAAC
Special Heritage Status Awarded by UGC
Tiruchirappalli - 620 002

Academic Year 2026 – 2027

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CERTIFICATE

This is to certify that the report on the Mini Project entitled **Job Vacancy Registration Portal** submitted to the Department of Computer Science, St. Joseph's College (Autonomous), Tiruchirappalli - 620 002, in partial fulfillment for the award of the Degree of **Bachelor of Science in Computer Science**, is an original work done by the following students under my supervision and guidance during the academic year **2026 - 2027**:

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CERTIFICATE OF SUSTAINABLE DEVELOPMENT GOALS (SDG) COMPLIANCE

This is to certify that the Mini Project entitled **Job Vacancy Registration Portal** has been done in alignment with the United Nations Sustainable Development Goals (SDGs) by the following students of the Department of Computer Science, St. Joseph's College (Autonomous), Tiruchirappalli – 620 002, in partial fulfillment of the requirements for the award of the Degree of Bachelor of Science in Computer Science.

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This Mini Project was carried out under my supervision and guidance during the academic period 2026 - 2027. The work is found to be original and reflects the students' ability to apply sustainability principles in software development.

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DECLARATION

We hereby declare that the Mini Project Report entitled **Job Vacancy Registration Portal** is a record of original work done by us during the period of study, under the guidance of **Mrs. M. Nasreen, M.C.A., M.Phil. and Dr. A. Janet Rajeswari, M.Sc., M.Phil., Ph.D.** Department of Computer Science, St. Joseph's College, Tiruchirappalli – 620 002.

We further declare that any part of this Mini Project has not been submitted elsewhere for the award of any Degree or Diploma at any other University or Research Institute.

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ACKNOWLEDGEMENT

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We are deeply indebted to our project guides, Mrs. M. Nasreen, M.C.A., M.Phil., and Dr. A. Janet Rajeswari, M.Sc., M.Phil., Ph.D., for their valuable guidance, constant support and suggestions during the development of the Job Vacancy Registration Portal.

We also thank all the faculty members of the Department of Computer Science, our parents and our friends, whose motivation and help made this project possible.

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ABSTRACT

The Job Vacancy Registration Portal is a web-based system developed for the recruitment portal of TalentBridge Solutions to let companies register their job openings online. In the traditional method a company posts a requirement through phone calls, e-mails or hand-written forms. The details are scattered, often incomplete, and difficult for a placement or recruitment team to verify, search and store. The portal replaces this with a single, structured online workflow.

A user first signs in with a register number or D.No as the username and the date of birth as the password. After a successful login the Job Vacancy Registration form opens. The form is divided into three sections: Company details, Job details and Hiring process. It collects the company name, official e-mail, contact number, website, industry type, company size, address, city, PIN code and optional GSTIN, followed by the job title, department, job type, work mode, location, number of openings, experience, job level, salary range, qualification, last date to apply, skills and description, and finally the application e-mail, expected joining time, interview mode, work shift, benefits and extra instructions. A unique Vacancy ID and the posting date are generated automatically.

Every field is checked in the browser before submission. Formats such as the 10-digit mobile number, the 6-digit PIN code, the 15-character GSTIN, the e-mail address and the website URL are enforced, the maximum salary cannot be lower than the minimum salary, the last date to apply cannot be in the past, and the job description must contain at least 50 characters. When all rules are satisfied the vacancy is registered, a success message is shown and the record is stored in the table job_vacancies of the MySQL database job_portal.

The front end is built with HTML, CSS and JavaScript. The application runs on the Apache web server supplied with the XAMPP package, and the data is stored in a MySQL database administered through phpMyAdmin using SQL. The project reduces paperwork, avoids incomplete entries through validation, and gives the recruitment team a single database from which vacancies can be searched, counted and reported.

TABLE OF CONTENTS

S.NO.	TITLE	PAGE NO.
	Acknowledgement	**5**
	Abstract	**6**
1	**Introduction**	**10**
1.1	Overview	10
1.2	Problem Statement	10
1.3	Objectives	10
1.4	Scope of the Project	11
1.5	Organisation of the Report	11
2	**System Analysis**	**12**
2.1	Existing System	12
2.2	Proposed System	12
2.3	Feasibility Study	13
2.4	Requirement Analysis	13
2.5	Software Development Approach	13
3	**System Requirements**	**15**
3.1	Hardware Requirements	15
3.2	Software Requirements	15
4	**Technologies Used**	**16**
4.1	HTML (HyperText Markup Language)	16
4.2	CSS (Cascading Style Sheets)	16
4.3	JavaScript	17
4.4	XAMPP Server	17
4.5	MySQL, phpMyAdmin and SQL	18
4.6	Summary of Technologies	18
5	**System Design**	**19**
5.1	System Architecture	19
5.2	Modules of the System	19
5.3	Process Flow	20
5.4	Data Flow Diagrams	21
5.5	Database Design	22
5.6	Input Design	24
5.7	Validation Rules	26
5.8	User Interface Design	26
5.9	Security and Data Integrity	27
5.10	Server-side Processing	27
6	**Implementation**	**29**
6.1	Step 1: Starting the XAMPP Server	29
6.2	Step 2: Creating the Database	29
6.3	Step 3: Login Page	30
6.4	Step 4: Login Validation	30
6.5	Step 5: Form Header and Company Details	31
6.6	Step 6: Job Details	32
6.7	Step 7: Salary, Qualification and Description	33
6.8	Step 8: Hiring Process	34
6.9	Step 9: Confirmation and Registration	34

S.NO.	TITLE	PAGE NO.
6.10	Step 10: Verifying the Record in the Database	35
6.11	Step 11: Logout	35
7	**Testing**	**36**
7.1	Testing Approach	36
7.2	Test Cases	36
7.3	Test Observations	37
7.4	Sample Test Data	38
8	**Conclusion and Future Enhancement**	**39**
8.1	Conclusion	39
8.2	Limitations of the Present System	39
8.3	Future Enhancements	39
9	**Bibliography**	**41**

LIST OF TABLES

Table 2.1: Feasibility study....13
Table 2.2: Phases of development....14
Table 3.1: Hardware requirements....15
Table 3.2: Software requirements....15
Table 4.1: Summary of technologies....18
Table 5.1: Responsibilities of each tier....19
Table 5.2: Process flow of the portal....20
Table 5.3: Structure of the table job_vacancies....22
Table 5.4: Section 1 – Company details....24
Table 5.5: Section 2 – Job details....25
Table 5.6: Section 3 – Hiring process....25
Table 5.7: Validation rules and messages....26
Table 5.8: Files of the project....27
Table 7.1: Test cases and results....36
Table 7.2: Sample valid and invalid test data....38

LIST OF FIGURES

Figure 5.1: Three-tier architecture of the portal....19
Figure 5.2: Flowchart of the vacancy registration process....21
Figure 5.3: Data Flow Diagram – Level 0....22
Figure 5.4: Data Flow Diagram – Level 1....22
Figure 5.5: Entity Relationship (ER) diagram of the portal....24
Figure 6.1: XAMPP Control Panel with Apache and MySQL running....29
Figure 6.2: TalentBridge Solutions recruitment portal – login page....30
Figure 6.3: Job Vacancy Registration form – Company details....32
Figure 6.4: Job Vacancy Registration form – Job details....33
Figure 6.5: Job Vacancy Registration form – Hiring process and confirmation....34
Figure 6.6: phpMyAdmin showing the stored record in table job_vacancies....35
Figure 7.1: Table job_vacancies in phpMyAdmin after registering a vacancy....37

CHAPTER 1
INTRODUCTION

1.1 Overview

Recruitment is the process through which organisations find and appoint suitable people for their open positions. Before a candidate can be selected, the company has to announce the vacancy: it must describe the job, the skills and qualification required, the salary offered, the place of work, the last date to apply and the way in which applications should be sent. A recruitment portal such as TalentBridge Solutions acts as a bridge between companies that have openings and candidates who are looking for them.

This mini project, the Job Vacancy Registration Portal, automates the first stage of that bridge, which is the registration of a job vacancy by the company. It provides a login page with company branding, an online registration form arranged in three sections, automatic generation of a Vacancy ID and posting date, instant validation of every field, a success message after registration, and a MySQL database in which every registered vacancy is stored.

The portal is built as a simple three-tier web application. HTML, CSS and JavaScript form the front end, the Apache server of the XAMPP package hosts the pages, and MySQL, managed through phpMyAdmin, stores the data. All these tools are free and run on an ordinary Windows computer, which makes the project easy to set up in a college laboratory.

1.2 Problem Statement

In many small and medium organisations a job opening is communicated informally. The human resource officer sends an e-mail, calls the placement office or fills in a paper form. The recruitment team then has to collect these scattered messages and convert them into a proper vacancy record. The problems with this approach are:

- Important details such as salary range, experience required, last date to apply or work mode are frequently missing.
- Contact numbers, e-mail addresses and PIN codes are typed in wrong formats and cannot be used later.
- There is no uniform structure, so two vacancies from two companies cannot be compared or searched easily.
- Paper forms and e-mails can be lost or duplicated, and nothing tells the company that its vacancy has been recorded.
- Counting vacancies by industry, city or job type for a report requires manual work.

1.3 Objectives

The objectives of the project are:

- To provide a login-protected web portal so that only authorised users can register vacancies.

- To collect company details, job details and hiring-process details through a single, well-organised online form.
- To validate inputs such as e-mail, mobile number, PIN code, GSTIN, website, salary range and deadline before they are accepted.
- To generate a unique Vacancy ID and the posting date automatically so that the user does not type them.
- To give clear on-screen error messages and a success message after registration.
- To store every registered vacancy permanently in a MySQL database that can be queried using SQL.

1.4 Scope of the Project

The present version covers the employer side of the recruitment process: login, vacancy registration form, validation, success message, storage of the record and logout, together with the database that holds the vacancies. It can be used by any company that wishes to post an opening through the portal. The scope can be extended later with an admin panel for the recruitment team, a candidate-side module for searching and applying for jobs, resume upload, e-mail notification and reporting dashboards.

1.5 Organisation of the Report

Chapter 2 studies the existing and proposed systems. Chapter 3 lists the hardware and software requirements. Chapter 4 explains the technologies used, namely HTML, CSS, JavaScript, the XAMPP server, MySQL and SQL. Chapter 5 describes the design of the system, its process flow, data flow diagrams, database and input rules. Chapter 6 explains the implementation step by step with screenshots of the running application. Chapter 7 presents testing, and Chapter 8 gives the conclusion and future enhancements. The sources used are listed in Chapter 9.

CHAPTER 2
SYSTEM ANALYSIS

2.1 Existing System

At present a company that wants to announce a job opening usually sends its requirement as a free-text e-mail, a phone message or a printed letter. The recruitment staff read the message, ask for the missing details, copy the information into a register or a spreadsheet and then publish the vacancy. Any mistake or missing item means another round of communication with the company.

Limitations of the existing system:

- Time-consuming and dependent on office hours and on the availability of the staff.
- High chance of missing, wrong or inconsistent information such as salary, experience or deadline.
- No standard format, so vacancies cannot be searched, filtered or counted quickly.
- No instant acknowledgement or reference number is given to the company.
- Records kept in papers, e-mails and personal spreadsheets are difficult to secure and back up.

2.2 Proposed System

The proposed system is a web portal that can be opened in a browser. The recruiter logs in, fills in one structured form and registers the vacancy. Mandatory fields and format rules are checked by the form itself, so incomplete or wrongly formatted data cannot be submitted. Fields such as the Vacancy ID and the posting date are produced by the system and are read-only. When registration is completed, a success message is displayed and the vacancy is held in the database.

Advantages of the proposed system:

- Paperless, available at any time and from any computer that can reach the server.
- Reduces errors through required-field checks, format rules and read-only generated fields.
- Provides an instant success message and a unique Vacancy ID for every posting.
- Stores all vacancies in one database so that they can be searched, counted and exported using SQL.
- Easy to extend with an administrator module, candidate module and reports.

2.3 Feasibility Study

Table 2.1: Feasibility study

Feasibility	Observation
Technical	The system uses HTML, CSS, JavaScript, Apache and MySQL, all of which are free, well documented and run on an ordinary Windows computer through XAMPP.
Economic	No licence cost is involved because every tool used is open source. Only the existing college computers and network are required.
Operational	The interface is simple, uses labelled fields with examples and hints, and shows clear error messages, so users can work with it without special training.
Schedule	The project is small and modular, so the design, coding, testing and documentation could be completed within the semester.

2.4 Requirement Analysis

Functional requirements

- The system shall allow a user to log in using a register number or D.No and a date of birth.
- The system shall display the vacancy form only after a successful login and shall allow logout at any time.
- The system shall collect company, job and hiring details as listed in Chapter 5.
- The system shall generate the Vacancy ID and the posting date automatically.
- The system shall reject the form until every rule is satisfied and shall point to the field in error.
- The system shall store each registered vacancy as one record in the database.

Non-functional requirements

- Usability: a clean two-column layout, hints below fields, visible required-field marks and examples in every box.
- Performance: validation happens instantly in the browser, and a stored record is retrieved in a fraction of a second, as seen in the phpMyAdmin screenshot.
- Portability: runs on any modern browser such as Chrome, Edge or Firefox on a standard screen.
- Responsiveness: the layout adapts from a wide desktop screen to a narrow mobile screen.
- Maintainability: a modular structure with separate login, form, and database parts.

2.5 Software Development Approach

The project was developed with an iterative approach. The requirements were first fixed from the fields that a recruiter needs to announce a job, then the screens were designed and coded,

and each part was tested before the next one was added. Working in small steps made it easy to correct mistakes early and to add improvements such as the automatic Vacancy ID and the salary check.

Table 2.2: Phases of development

Phase	Work carried out
Requirement gathering	Listing the company, job and hiring details needed for a complete vacancy and the rules each field must follow
Design	Planning the login screen, the three-section form, the colour scheme, the database table and the process flow
Coding	Writing the HTML structure, CSS styling and JavaScript behaviour, and configuring XAMPP and MySQL
Testing	Trying valid and invalid inputs for every field and checking the stored record in phpMyAdmin
Documentation	Preparing this report with screenshots, diagrams and test results

CHAPTER 3
SYSTEM REQUIREMENTS

3.1 Hardware Requirements

Table 3.1: Hardware requirements

Component	Requirement
Processor	Intel Core i3 or higher (or equivalent)
RAM	4 GB or more
Hard disk	500 MB free space for XAMPP and project files
Display	1366 x 768 resolution or higher
Network	Local network or internet for accessing the portal from other computers

3.2 Software Requirements

Table 3.2: Software requirements

Software	Details
Operating system	Windows 64-bit (the project was developed on Windows Home 64-bit)
Web server package	XAMPP version 8.2.12 with Control Panel v3.3.0 (Apache and MySQL)
Web server ports	Apache on ports 8080 and 443, MySQL on port 3306
Database	MySQL, administered through phpMyAdmin; SQL used for all queries
Front-end languages	HTML, CSS and JavaScript
Web browser	Google Chrome, Microsoft Edge or Mozilla Firefox (latest version)
Project files	index.html (login and form), submit_job.php (saves the form), db.php (database connection) and database.sql (creates the database and table), kept in the folder job_vacancy_xampp
Other tools	A text or code editor

The project folder job_vacancy_xampp is placed inside the htdocs directory of the XAMPP installation (C:\xampp\htdocs). Files placed in this directory are served by Apache, so the portal can be opened in the browser through the local server address.

CHAPTER 4
TECHNOLOGIES USED

The portal is built on a standard three-layer web stack: a browser-based front end written in HTML, CSS and JavaScript, a web server provided by XAMPP, and a SQL database. Each technology is explained below together with the way it is used in this project.

4.1 HTML (HyperText Markup Language)

HTML is the standard markup language for creating web pages. It describes the structure of a page using elements such as headings, paragraphs, images, links and form controls. In this project HTML builds the two screens of the portal: the split-screen login page and the Job Vacancy Registration form.

Key HTML features used in the project:

- Form controls: text boxes, e-mail, telephone, URL and number inputs, drop-down lists (industry, company size, department, work mode, job level, qualification, expected joining, work shift), radio buttons (job type and interview mode), check boxes (benefits and the final confirmation), text areas (job description and extra instructions) and a date picker (last date to apply).
- Grouping elements: each section of the form is a fieldset with a legend, namely Company details, Job details and Hiring process.
- Attributes such as required, minlength, maxlength, min, max, step, pattern, placeholder, title and readonly, which guide the user and restrict the input.
- Inline vector graphics: the TalentBridge logo and the head-office building shown on the login screen are drawn with scalable vector graphics, so no image file is needed.

4.2 CSS (Cascading Style Sheets)

CSS controls the look and layout of HTML pages. It is used here to give the portal a uniform, professional design that matches the TalentBridge brand.

- A navy, teal and white colour scheme defined once as colour variables and reused across the page; green is used for success, and red for errors.
- A white rounded card with a soft shadow holds the form, with a dark navy header bar carrying the title and the Log out button.
- A split-screen login layout in a 16:9 shape: a branded gradient panel with the company tagline on the left and the sign-in form on the right.
- Each section heading has a teal underline, so the long form is easy to scan.
- Grid layout places two fields side by side, and media queries stack them in a single column on narrow screens.

- Validation states: a field turns red with a light red background when its content is wrong and green when it is correct, and the matching error message appears below the field only after the user has interacted with it.
- Read-only boxes, such as the Vacancy ID and Posted on fields, are shown in a pale grey so that the user knows they cannot be edited.

4.3 JavaScript

JavaScript adds behaviour and interactivity to the pages in the browser. In this project it is used for:

- Switching between the login screen and the registration form after a successful login, and back again when the user logs out.
- Showing or hiding the password when the Show button is pressed; the label of the button changes between Show and Hide.
- Generating the Vacancy ID in the form TB-year-number and filling the Posted on field with the current date in the Indian date format.
- Making sure that the maximum salary cannot be set lower than the minimum salary by updating the allowed minimum of the maximum-salary field as the user types.
- Converting the GSTIN to capital letters automatically while it is typed.
- Blocking past dates in the Last date to apply field by setting the minimum selectable date to today.
- Running the validation of the whole form when the Register vacancy button is pressed, so that only a form that satisfies every rule is sent to the server.

4.4 XAMPP Server

XAMPP is a free package from Apache Friends that installs everything needed to run a web application on a local computer. The name stands for Cross-platform, Apache, MariaDB/MySQL, PHP and Perl. In this project XAMPP version 8.2.12 is used.

- Apache HTTP Server: receives the browser’s requests and serves the HTML, CSS and JavaScript of the portal. In this installation it listens on ports 8080 and 443.
- MySQL database server: stores the vacancy data. It runs on port 3306.
- PHP: the server-side scripting language bundled with XAMPP, used to receive the form data and exchange it with the database. Two PHP files are used here: db.php opens the connection to MySQL and submit_job.php receives the form and saves it.
- phpMyAdmin: a web tool for creating databases and tables and for browsing and running SQL queries.
- XAMPP Control Panel: used to start and stop the Apache and MySQL modules and to read their log messages.

4.5 MySQL, phpMyAdmin and SQL

MySQL is an open-source relational database management system. Data is stored in tables made of rows and columns, and every row is identified by a primary key. SQL (Structured Query Language) is the standard language for creating and managing relational databases. In the project it is used to create the database job_portal and the table job_vacancies, to insert each registered vacancy (INSERT), to read records (SELECT) and, when needed, to correct or remove records (UPDATE and DELETE).

phpMyAdmin is the browser-based administration tool for MySQL. It is opened from the Admin button next to MySQL in the XAMPP Control Panel. It shows the databases in a tree on the left, lets the developer browse the table, run SQL statements and view the structure of the table, as shown in Chapter 6.

4.6 Summary of Technologies

Table 4.1: Summary of technologies

Layer	Technology	Purpose in the project
Presentation	HTML	Structure of the login page and the vacancy form
Presentation	CSS	Design, colours, layout, validation colours and responsiveness
Behaviour	JavaScript	Screen switching, password toggle, Vacancy ID, date rules, validation
Server	XAMPP (Apache, PHP)	Hosts the portal and processes form data
Database	MySQL / phpMyAdmin	Stores vacancies in job_portal
Query language	SQL	Creating tables and querying data

CHAPTER 5
SYSTEM DESIGN

5.1 System Architecture

The portal follows a three-tier architecture. The user works in a web browser (client tier). The browser sends requests to the Apache server running under XAMPP (application tier), which processes the form data and talks to the MySQL database (data tier). The results are returned to the browser as pages and messages such as “Vacancy registered successfully”.

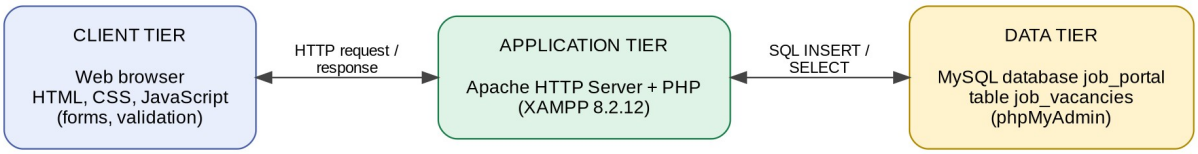


Figure 5.1: Three-tier architecture of the portal

Table 5.1: Responsibilities of each tier

Tier	Component	Responsibility
Client	Browser with HTML, CSS, JavaScript	Displays pages, validates input, shows messages
Application	Apache server + PHP (submit_job.php, db.php)	Receives the form data, cleans it and saves it in the database
Data	MySQL database job_portal	Stores registered vacancies permanently

5.2 Modules of the System

- Authentication module: split-screen login page, password show/hide, validation of the username and password formats, and logout.
- Company details module: collects the identity, contact, industry, size and address details of the company.
- Job details module: collects the job title, department, type, work mode, location, openings, experience, level, salary, qualification, deadline, skills and description, and generates the Vacancy ID and posting date.
- Hiring process module: collects the application e-mail, joining time, interview mode, work shift, benefits and extra instructions.
- Validation module: checks every rule and displays messages next to the fields.
- Server-side module: submit_job.php receives the posted form and db.php connects PHP to MySQL.
- Database module: stores each registered vacancy as a row in the table job_vacancies.

5.3 Process Flow

Table 5.2: Process flow of the portal

Step	Process	Result
1	Start Apache and MySQL in XAMPP	Server and database are running
2	Open the portal in the browser	Login screen is displayed
3	Enter register number and date of birth	Valid: form opens. Invalid: message is shown
4	Fill Company, Job and Hiring details	All sections completed
5	Tick the confirmation and click Register vacancy	Validation runs; success message appears
6	View the record in phpMyAdmin	One row added in table job_vacancies
7	Click Log out	Session ends and user returns to the login screen

The flowchart in Figure 5.2 shows the same process graphically. It starts at the login screen, checks the entered values, opens the registration form on success, validates the form and registers the vacancy only when every rule is satisfied.

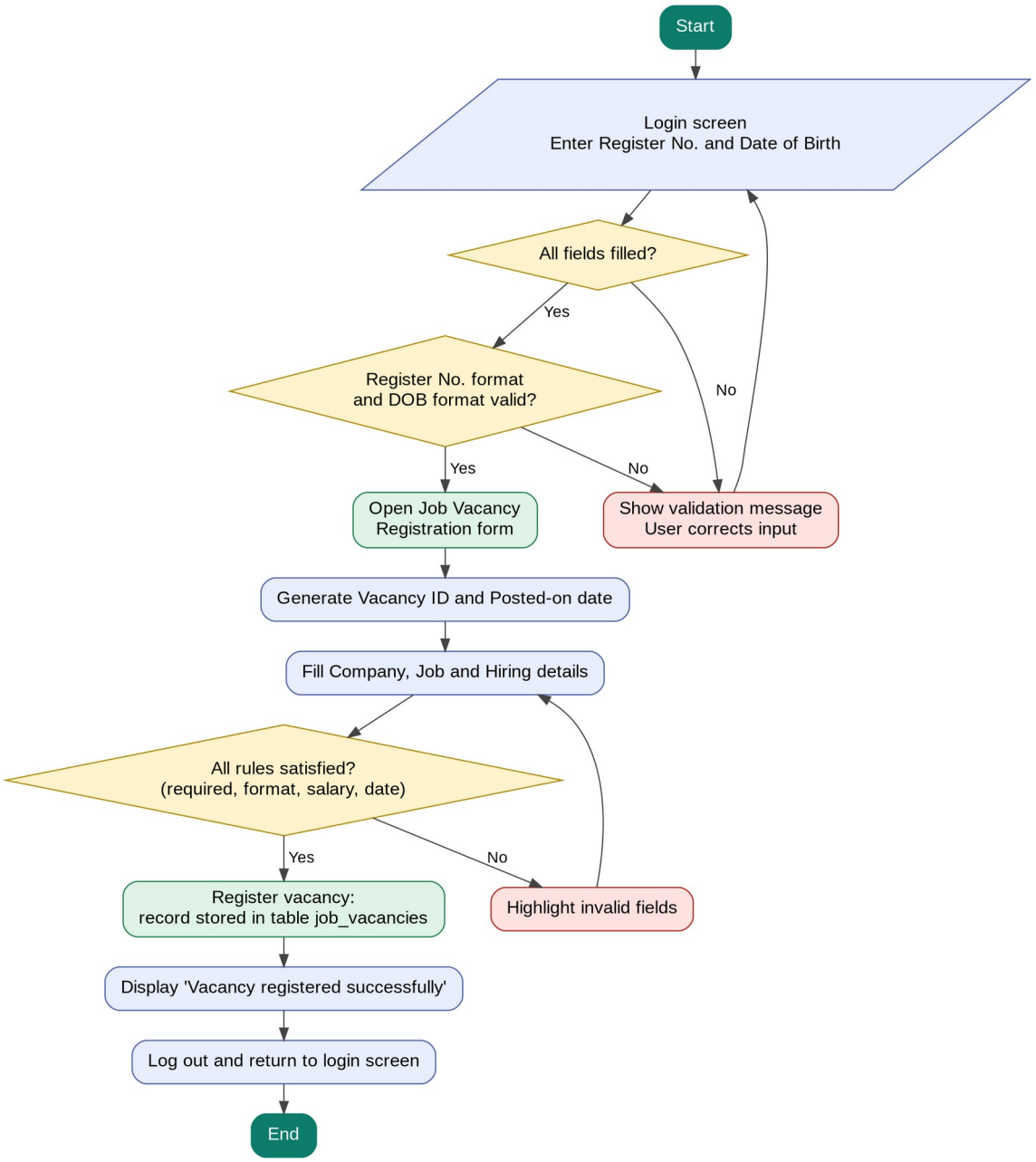


Figure 5.2: Flowchart of the vacancy registration process

5.4 Data Flow Diagrams

A Data Flow Diagram (DFD) shows how data moves between the user, the processes of the system and the data store. The context-level diagram (Level 0) in Figure 5.3 shows the whole portal as a single process with one external entity, the employer or recruiter, and one data store, the database job_portal.

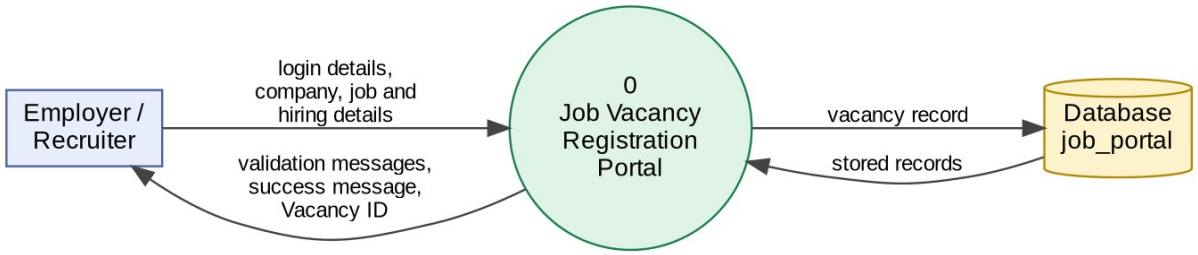


Figure 5.3: Data Flow Diagram – Level 0

Figure 5.4 breaks the portal into four processes. Process 1.0 authenticates the user. Process 2.0 collects the form entries and validates them. Process 3.0 supplies the generated Vacancy ID and posting date. Process 4.0 stores the valid data in the table job_vacancies and returns the success message.

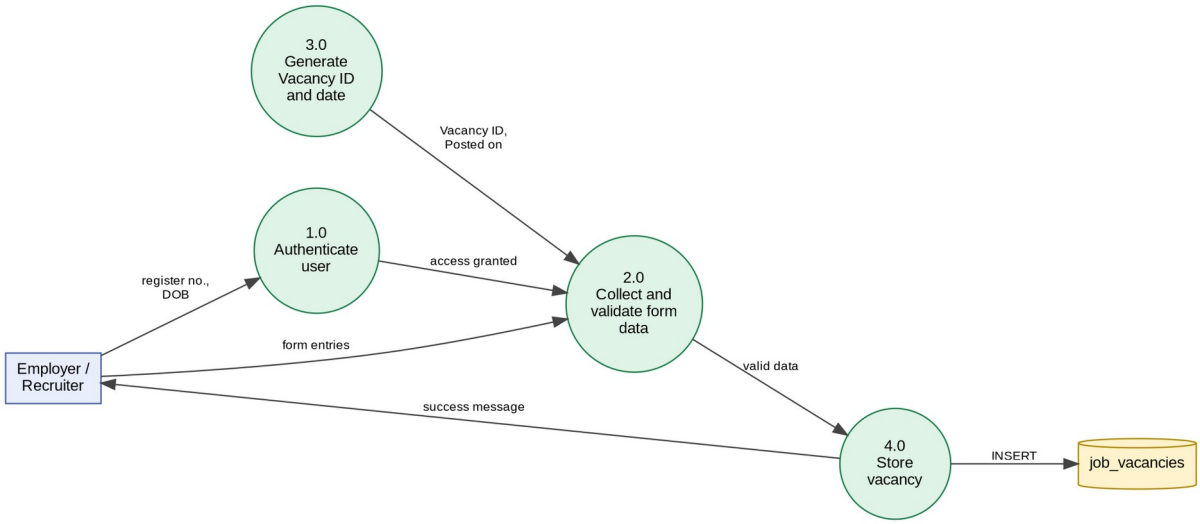


Figure 5.4: Data Flow Diagram – Level 1

5.5 Database Design

The database is named job_portal and the main table is job_vacancies. Each registered vacancy is stored as one row. Both are created by the script file database.sql, which is run once in the SQL tab of phpMyAdmin. Columns marked required cannot be left empty in the database. The columns of the table are listed below.

Table 5.3: Structure of the table job_vacancies

Column	Type	Description
id	INT, primary key, auto-increment	Unique number of the row
company	VARCHAR(80), required	Name of the company
email	VARCHAR(100), required	Official e-mail address
phone	VARCHAR(10), required	Contact number (10 digits)
contact	VARCHAR(50)	Contact person (optional)
website	VARCHAR(150)	Company website (optional)

Column	Type	Description
industry	VARCHAR(50), required	Industry type
size	VARCHAR(20), required	Company size range
address	VARCHAR(150), required	Company address
city	VARCHAR(40), required	City
pincode	VARCHAR(6), required	6-digit PIN code
gstin	VARCHAR(15)	GST number (optional)
vacid	VARCHAR(20)	Vacancy ID, e.g. TB-2026-2908
posted	VARCHAR(20)	Posting date shown on the form
title	VARCHAR(60), required	Job title
dept	VARCHAR(50), required	Department
jobtype	VARCHAR(20), required	Full-time, Part-time, Internship or Contract
mode	VARCHAR(20), required	On-site, Hybrid or Remote
location	VARCHAR(60), required	Job location
openings	INT, required	Number of openings
exp	INT, required	Experience required in years
level	VARCHAR(30), required	Job level
salmin	INT, required	Minimum salary per year (INR)
salmax	INT, required	Maximum salary per year (INR)
qualification	VARCHAR(30), required	Minimum qualification
deadline	DATE, required	Last date to apply
skills	VARCHAR(150), required	Skills required
description	TEXT, required	Job description
applymail	VARCHAR(100), required	E-mail to which applications are sent
joining	VARCHAR(30), required	Expected joining time
interview	VARCHAR(20), required	Interview mode
shift	VARCHAR(20), required	Work shift
benefits	VARCHAR(255)	Ticked benefits, separated by commas
notes	VARCHAR(300)	How to apply / extra instructions
created_at	TIMESTAMP, default current time	Date and time at which the row was saved automatically

The Entity Relationship (ER) diagram in Figure 5.5 shows the logical data model of the portal. The data of the form can be grouped into the entities Company, Vacancy, Hiring Process and Benefit. One company can post many vacancies, each vacancy follows one hiring process and can offer many benefits. In the present version these entities are stored together as a single row

of the table job_vacancies, which keeps the first version simple; the logical model shows how the data can be normalised when the system grows.

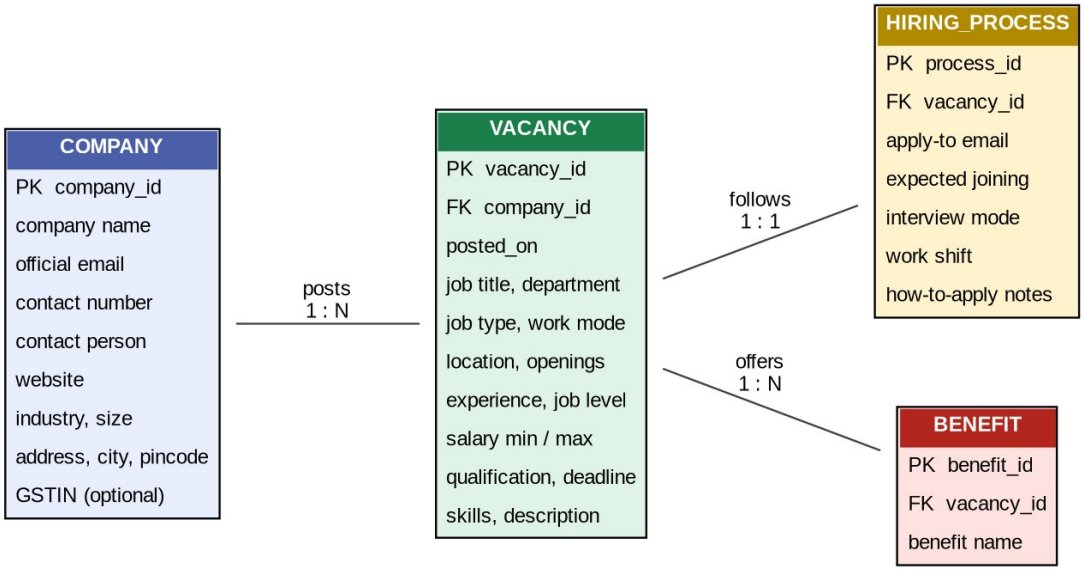


Figure 5.5: Entity Relationship (ER) diagram of the portal

5.6 Input Design

The Job Vacancy Registration form is divided into three sections so that the long form is easy to complete. Required fields are marked with a red asterisk, and hints are shown below the labels.

Table 5.4: Section 1 – Company details

Field	Type	Rule
Company name *	Text	2 to 80 characters; letters, digits and basic punctuation
Official email *	E-mail	Valid e-mail address, e.g. hr@company.com
Contact number *	Telephone	10 digits, first digit 6, 7, 8 or 9
Contact person	Text (optional)	3 to 50 letters, spaces and dots
Company website	URL (optional)	Full address starting with http:// or https://
Industry type *	Drop-down	IT and Software, Banking and Finance, Manufacturing, Healthcare, Education, Retail and E-commerce, Logistics and Transport, Other
Company size *	Drop-down	1 - 10, 11 - 50, 51 - 200, 201 - 1000, 1000+
Company address *	Text	10 to 150 characters
City *	Text	2 to 40 letters, spaces, dots and hyphens
PIN code *	Text	6 digits, not starting with 0
GSTIN	Text (optional)	15-character GST number; converted to capitals

Table 5.5: Section 2 – Job details

Field	Type	Rule
Vacancy ID	Read-only	Generated automatically, e.g. TB-2026-2908
Posted on	Read-only	Current date, e.g. 05 Oct 2026
Job title *	Text	3 to 60 characters
Department *	Drop-down	Software Development, Data and Analytics, Networking and Security, Quality Assurance, Human Resources, Sales and Marketing, Other
Job type *	Radio buttons	Full-time, Part-time, Internship or Contract
Work mode *	Drop-down	On-site, Hybrid or Remote
Job location *	Text	2 to 60 characters
Number of openings *	Number	Whole number from 1 to 500
Experience required *	Number	0 to 40 years; 0 means freshers
Job level *	Drop-down	Entry level / Fresher, Junior, Mid level, Senior, Team lead / Manager
Minimum salary *	Number	Per year in INR, up to 1,00,00,000, in steps of 1000
Maximum salary *	Number	Not lower than the minimum salary
Minimum qualification *	Drop-down	Diploma, B.Sc / BCA, B.E / B.Tech, M.Sc / MCA, M.E / M.Tech, Any degree
Last date to apply *	Date	Today or a later date
Skills required *	Text	3 to 150 characters, separated by commas
Job description *	Text area	50 to 1000 characters

Table 5.6: Section 3 – Hiring process

Field	Type	Rule
Send applications to *	E-mail	Valid e-mail address, e.g. careers@company.com
Expected joining *	Drop-down	Immediate, within 15, 30, 60 or 90 days
Interview mode *	Radio buttons	Online, In-person or Telephonic
Work shift *	Drop-down	Day shift, Night shift, Rotational or Flexible
Benefits offered	Check boxes	Health insurance, Provident fund, Paid leave, Performance bonus, Work from home, Training
How to apply / instructions	Text area (optional)	Up to 300 characters
Confirmation *	Check box	Must be ticked to register the vacancy

5.7 Validation Rules

Validation protects the database from wrong or incomplete data. It is done in two ways. First, the HTML attributes of each field define the basic rules such as required, length, range and format. Second, JavaScript adds the rules that depend on other fields or on the current date.

Table 5.7: Validation rules and messages

Area	Rule	Message shown
Username	4 to 30 characters, required	Enter your register number, 4 to 30 characters.
Password (DOB)	Format DD-MM-YYYY	Enter your date of birth as DD-MM-YYYY, e.g. 15-08-2005.
E-mail	Valid e-mail format	Enter a valid email address, like hr@company.com.
Contact number	10 digits starting with 6 to 9	Enter a 10-digit number starting with 6, 7, 8 or 9.
PIN code	6 digits, first digit not 0	Enter a valid 6-digit PIN code.
GSTIN	15-character GST format	Enter a valid 15-character GSTIN.
Website	Starts with http:// or https://	Enter a full URL starting with http:// or https://.
Openings	Whole number 1 to 500	Enter a whole number between 1 and 500.
Salary	Maximum not below minimum	Enter an amount up to 1,00,00,000 that is not below the minimum salary.
Last date	Not before today	Choose a date that is today or later.
Description	At least 50 characters	Description must be at least 50 characters long.

5.8 User Interface Design

The interface was designed so that a first-time user can complete the form without any training. The following guidelines were followed:

- Consistency: the same colours, spacing and font are used on the login screen and on every section of the form.
- Clarity: every field has a visible label, a required-field mark where needed, an example as a placeholder and a short hint below the box where the format is not obvious.
- Grouping: related fields are placed together under a section heading with an underline, so the user always knows which part of the form is being filled.
- Feedback: a box turns green when its value is correct and red with a message when it is wrong, and a green banner confirms a successful registration.
- Protection: generated and fixed values such as the Vacancy ID and the posting date are shown in grey read-only boxes.

- Accessibility: labels are linked to their boxes, radio buttons and check boxes are grouped with names, and the focus of a box is shown with a coloured outline.

5.9 Security and Data Integrity

A recruitment portal handles contact details of companies, so basic protection of the data is important even in a college project. The present version provides the following measures, and stronger ones are listed as future work.

- Login gate: the registration form is not shown until the login values are entered in the required format.
- Input restriction: length limits, number ranges, patterns and drop-down lists keep unexpected characters and values out of the database.
- Read-only fields: the Vacancy ID and the posting date cannot be edited by the user.
- Primary key: every row of the table has a unique auto-incremented id, so a record can always be identified without confusion.
- Local deployment: the database runs on the local MySQL server and is reached only through the XAMPP installation of the computer.
- Prepared statement: the server saves the data with a prepared statement, in which the values travel separately from the SQL command, which protects the database against SQL injection.
- Cleaning of input: every value is trimmed and special characters are escaped before it is used.
- POST-only access: the saving file accepts only data sent by the form; if it is opened directly in the browser, the user is sent back to the form page.

5.10 Server-side Processing

Besides the page that the user sees, the portal has three supporting files in the project folder. They are described below in words; their code is not reproduced in this report.

Table 5.8: Files of the project

File	Purpose
index.html	Contains the login screen and the Job Vacancy Registration form with their styling and browser-side validation
database.sql	Script run once in phpMyAdmin to create the database job_portal and the table job_vacancies
db.php	Opens the connection from PHP to the MySQL server of XAMPP using the local host, the default user root and the database job_portal; stops with a message if the connection fails
submit_job.php	Receives the form data, cleans every value, joins the ticked benefits into one text, inserts one row into job_vacancies and sends the user back to the form page with a success or error indication

When the user presses Register vacancy and every rule is satisfied, the browser sends the form to submit_job.php using the POST method. The file first loads db.php to connect to the database, then reads each field of the form. Optional fields and unticked check boxes that are missing from the request are treated as empty text. The numeric fields (number of openings, experience and the two salaries) are converted to whole numbers. The ticked benefits are joined into a single text separated by commas. Finally the values are inserted into the table as one new row, and the browser is redirected to index.html with an indication of success, or with the error text if the insertion fails.

CHAPTER 6
IMPLEMENTATION

This chapter explains the working of the project step by step, in the same order in which a user and the developer use the system. Every step is supported by a screenshot of the running application.

6.1 Step 1: Starting the XAMPP Server

The portal runs on a local web server, so the first step is to start the required XAMPP modules.

- Open the XAMPP Control Panel (v3.3.0) from the installation folder c:\xampp\.
- Click the Start button next to Apache. The module name turns green, the process IDs (9212 and 7952) are shown and the ports 443 and 8080 are assigned.
- Click the Start button next to MySQL. It turns green with process ID 18704 and port 3306.
- Check the log window. It shows “Status change detected: running” for both Apache (07:39:06) and MySQL (07:39:16), which confirms that the server and the database are ready.

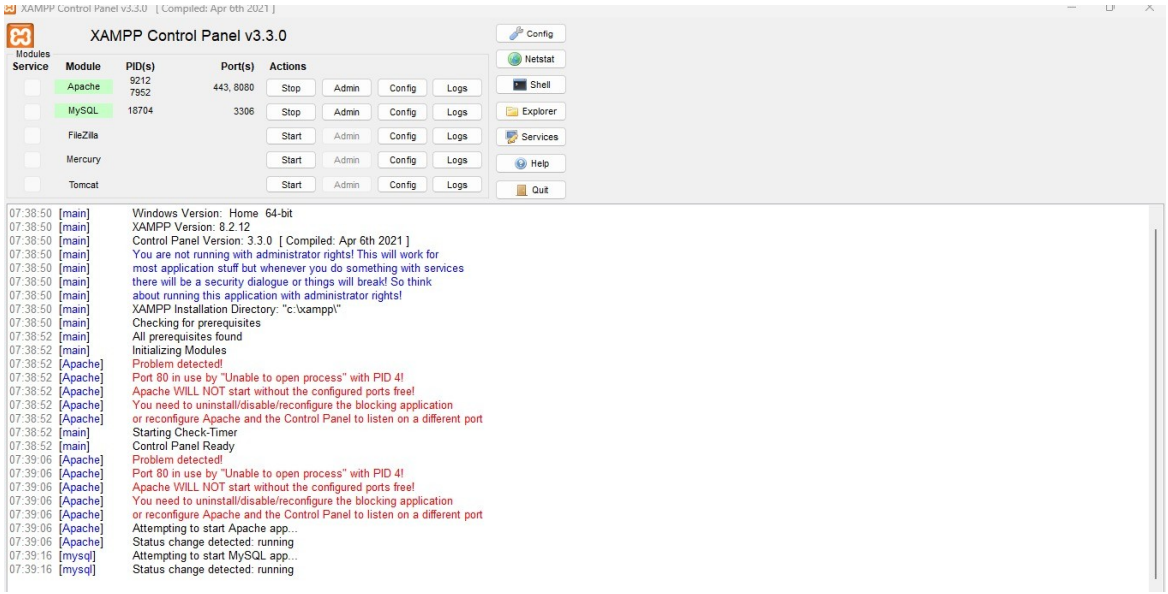


Figure 6.1: XAMPP Control Panel with Apache and MySQL running

The log also contains notices. The first says that the panel is not running with administrator rights, which is harmless for this project because the modules are started as normal applications. The second says that the default port 80 is already used by another process of the system (PID 4). Because of this, Apache was set to listen on port 8080 instead, and the portal and phpMyAdmin are therefore opened with the address localhost:8080 in the browser.

6.2 Step 2: Creating the Database

After MySQL is running, the Admin button next to MySQL opens phpMyAdmin. The database and the table are created with the script file database.sql kept in the project folder:

- Open the SQL tab of phpMyAdmin and run the script. It creates the database job_portal if it does not already exist and selects it.
- The script then creates the table job_vacancies with the columns listed in Table 5.3, covering the company details, job details and hiring process of the form.
- The column id is the primary key with auto-increment, so every row gets a unique number.
- The column created_at is filled automatically with the date and time at which a row is saved.

The left panel of phpMyAdmin then lists the databases information_schema, job_portal, mysql, performance_schema, phpmyadmin and test, and the node job_portal contains the table job_vacancies. The stored record is displayed later in Section 6.10.

6.3 Step 3: Login Page

When the user opens the portal, the first screen is the TalentBridge Solutions Recruitment Portal login page. It is a split screen in a 16:9 shape.

- The left panel carries the TalentBridge logo and name, the heading “Connecting great companies with great people.”, the line “Post your openings, reach qualified candidates and hire faster.” and an illustration of the company head office.
- The right panel shows “Welcome back” and “Sign in to register a job vacancy.”
- The Username (Register / D.No) box has the example 24CSU123 as a placeholder.
- The Password (Date of birth) box expects the date in the form DD-MM-YYYY and has a Show button, written in JavaScript, which reveals or hides the typed password.
- The green Log in button submits the entered values for checking.

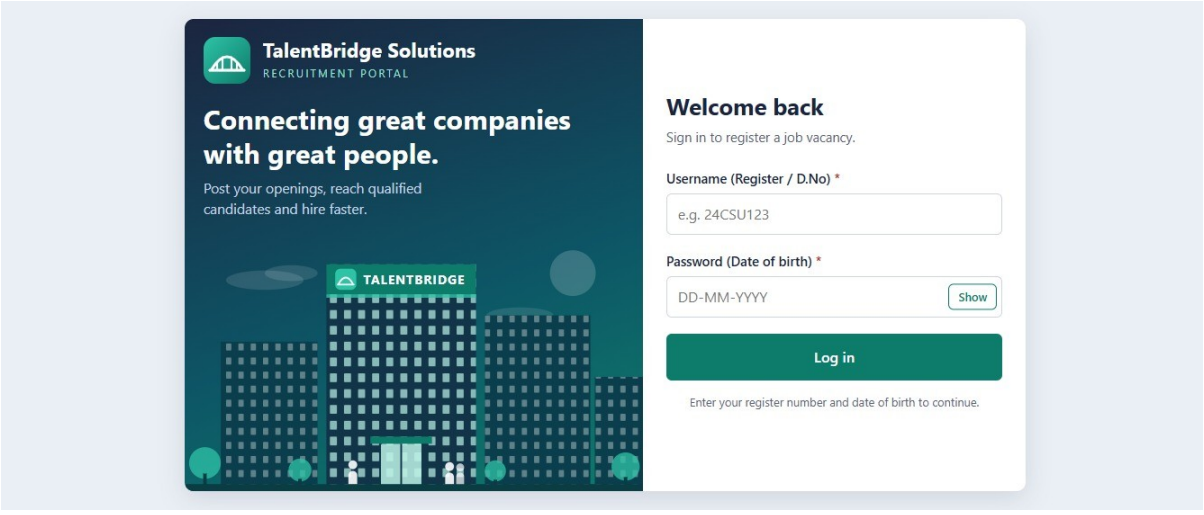


Figure 6.2: TalentBridge Solutions recruitment portal – login page

6.4 Step 4: Login Validation

When the Log in button is pressed, the page checks the two values. The username must be present and contain 4 to 30 characters. The password must follow the format DD-MM-YYYY,

for example 15-08-2005. If either value does not satisfy the rule, the border of the box turns red and a message such as “Enter your date of birth as DD-MM-YYYY” appears below it. The user cannot reach the form until both values are correct.

When the values are correct, the login screen is hidden and the Job Vacancy Registration form is displayed at the top of the page. The present version checks the format of the register number and the date of birth in the browser. Matching them against stored student records on the server is listed among the future enhancements in Chapter 8.

6.5 Step 5: Form Header and Company Details

After login the Job Vacancy Registration form opens. A navy header bar shows the title “Job Vacancy Registration”, the instruction “Fill in the details below to post an opening. Fields marked * are required.” and a Log out button at the right. The first section is Company details, which collects:

- Company name, official e-mail and contact number (10-digit mobile number, hint shown below the box).
- Contact person and company website (both optional).
- Industry type and company size, selected from drop-down lists.
- Company address, city and PIN code (6 digits).
- GSTIN, an optional 15-character GST number that is converted into capital letters as it is typed.

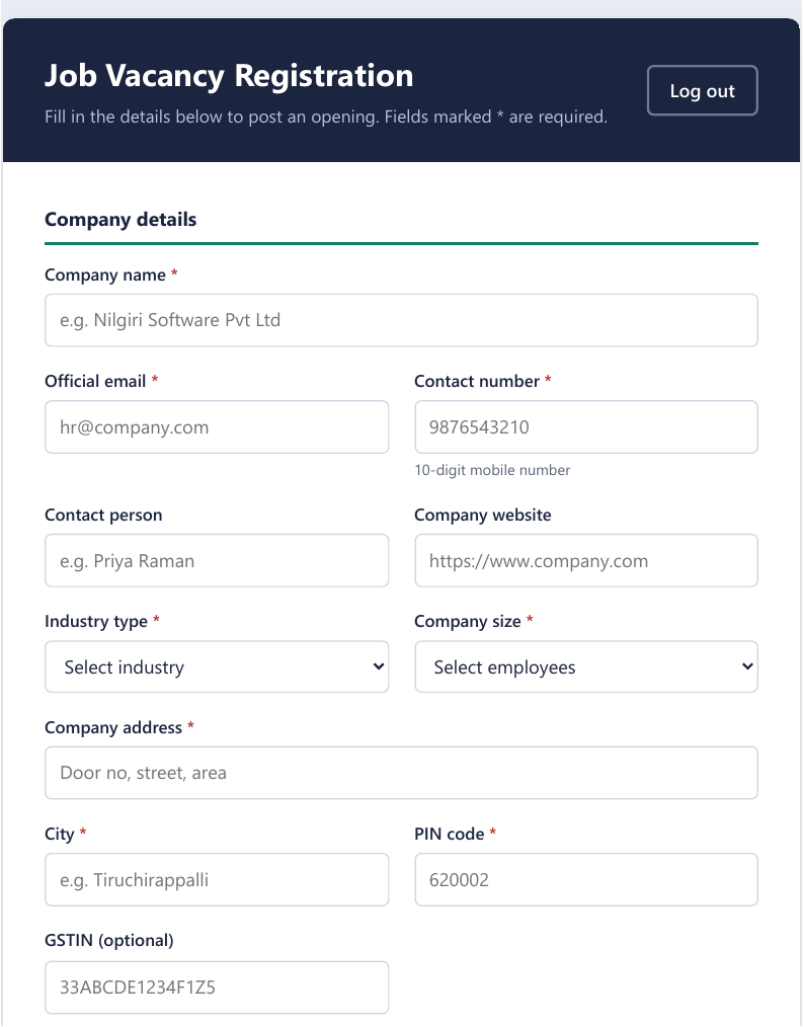


Figure 6.3: Job Vacancy Registration form – Company details

6.6 Step 6: Job Details

The Job details section starts with two grey read-only boxes. The Vacancy ID is generated automatically in the form TB-year-number (for example TB-2026-2908) and the Posted on box shows the current date (for example 05 Oct 2026). Because these are generated by the system, the user cannot change them. The user then enters:

- Job title and department.
- Job type as a radio button (Full-time, Part-time, Internship or Contract) and work mode as a drop-down (On-site, Hybrid or Remote).
- Job location and number of openings (a whole number between 1 and 500).
- Experience required in years (0 for freshers) and job level.

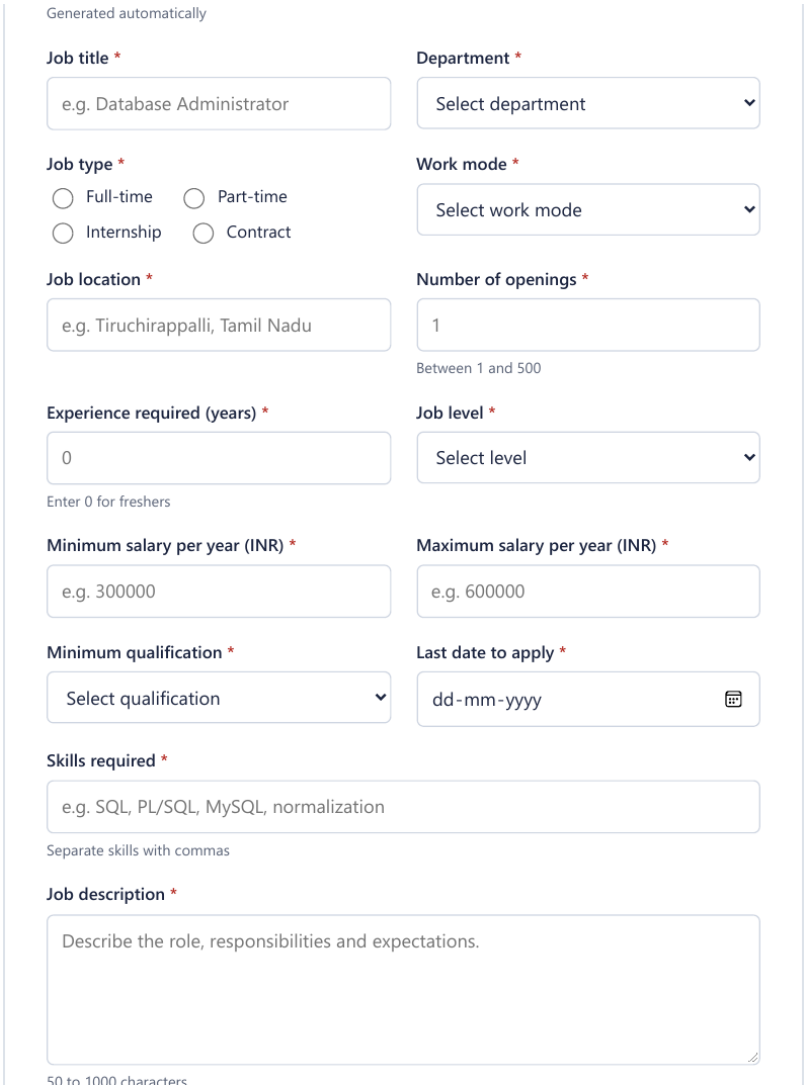


Figure 6.4: Job Vacancy Registration form – Job details

6.7 Step 7: Salary, Qualification and Description

The remaining job fields describe the pay and the expectations from the candidate.

- Minimum and maximum salary per year in INR. While the user types, the maximum is not allowed to be lower than the minimum.
- Minimum qualification, such as Diploma, B.Sc / BCA, B.E / B.Tech, M.Sc / MCA, M.E / M.Tech or Any degree.
- Last date to apply, selected with a calendar date picker in which past dates cannot be chosen.
- Skills required, entered as a comma-separated list, for example SQL, PL/SQL, MySQL, normalization.
- Job description of 50 to 1000 characters describing the role, responsibilities and expectations.

If any of these entries breaks a rule, the field turns red and the exact message is displayed below it, as listed in Table 5.7. When the entry is correct, the border turns green.

6.8 Step 8: Hiring Process

The Hiring process section records how the company will receive and process applications:

- Send applications to: the e-mail address that will receive the applications, such as careers@company.com.
- Expected joining: Immediate, within 15, 30, 60 or 90 days.
- Interview mode: Online, In-person or Telephonic.
- Work shift: Day shift, Night shift, Rotational or Flexible.
- Benefits offered: Health insurance, Provident fund, Paid leave, Performance bonus, Work from home and Training, any number of which can be ticked.
- How to apply / extra instructions: an optional note of up to 300 characters.

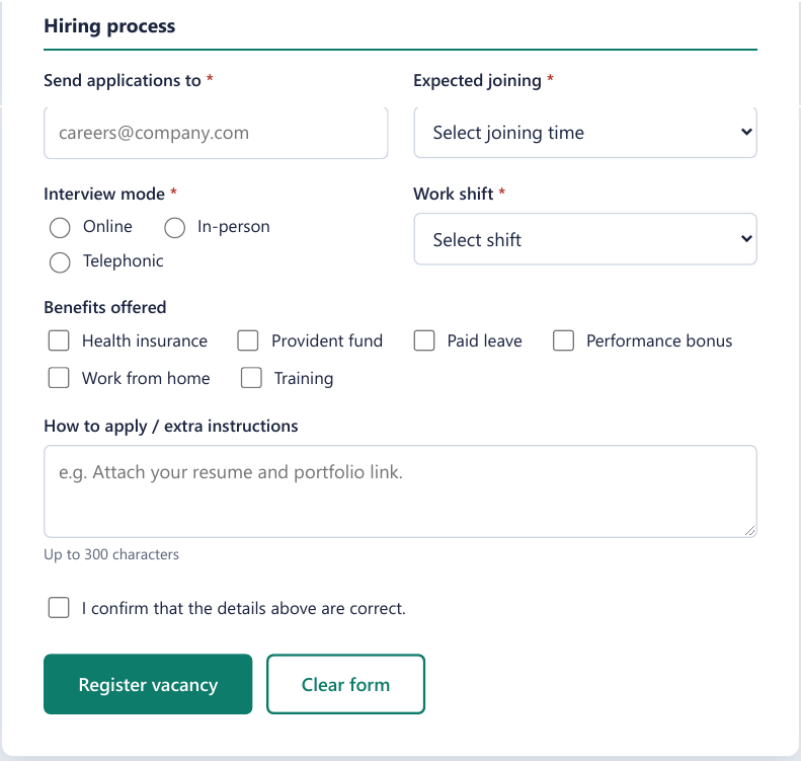


Figure 6.5: Job Vacancy Registration form – Hiring process and confirmation

6.9 Step 9: Confirmation and Registration

At the end of the form the user must tick the statement “I confirm that the details above are correct.” Two buttons are provided. The green Register vacancy button validates the whole form. If any rule is broken, the first wrong field is highlighted and the user is asked to correct it. When every rule is satisfied, the form data is sent by the POST method to the server-side file submit_job.php.

The file submit_job.php connects to the database through db.php, reads every field, removes extra spaces, escapes special characters, converts the numeric fields, joins the ticked benefits into one comma-separated text and inserts the data as one new row using a prepared statement. If the insertion succeeds, the browser is sent back to the form page with a success indication and

the message “Vacancy registered successfully.” is shown. If it fails, the form page is opened again with the error text. Opening submit_job.php directly, without submitting the form, simply returns the user to the form page.

The Clear form button empties all the boxes at once. After clearing, the Vacancy ID and the Posted on box are filled again with fresh values so that the form is ready for a new entry.

6.10 Step 10: Verifying the Record in the Database

To confirm that the data was stored, phpMyAdmin is opened for server 127.0.0.1 at the address localhost:8080/phpmyadmin, database job_portal, table job_vacancies. A SELECT query on the table displays the saved rows.

The result shows “Showing rows 0 - 0 (1 total, Query took 0.0006 seconds)”. The test record contains id 1, company marcello tech, industry IT and Software, company size 1 - 10, address near Chathiram Bus Stand, city Tiruchirappalli and PIN code 620002, together with the e-mail, phone, contact, website and the remaining fields. This proves that the data typed in the form reached the database correctly.

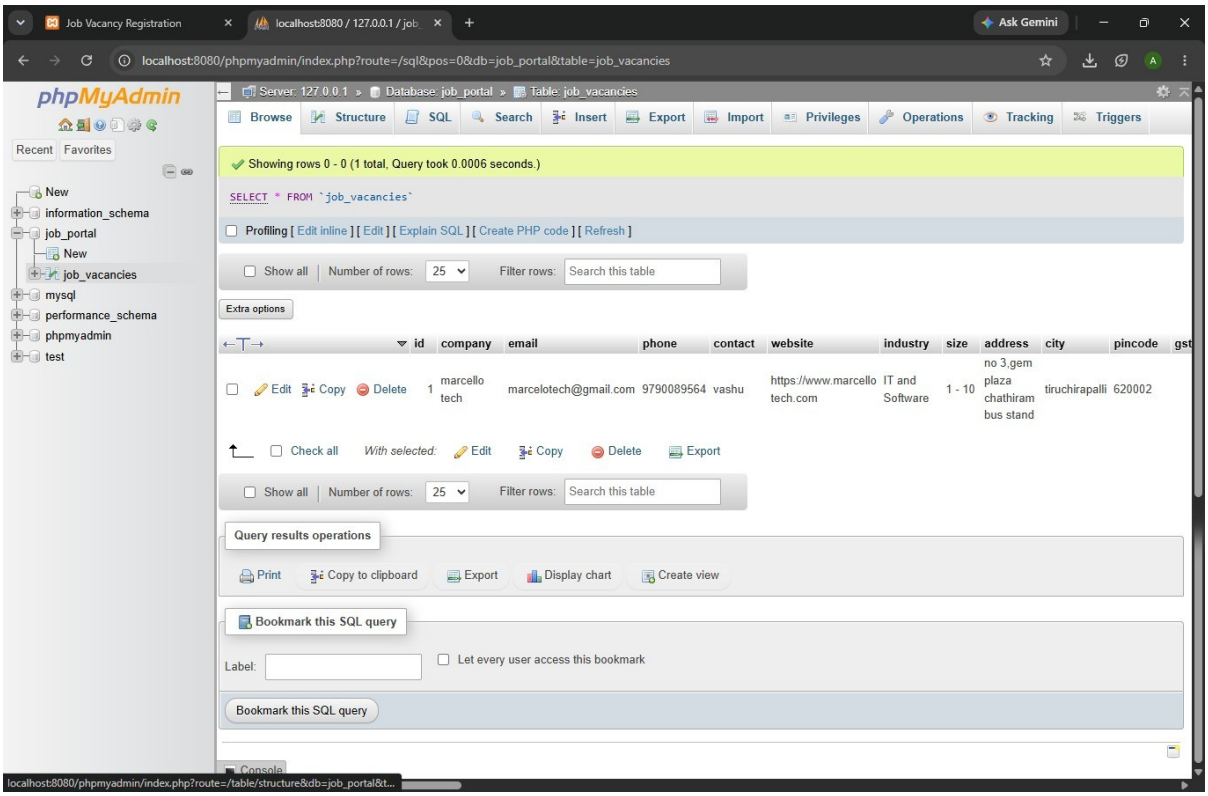


Figure 6.6: phpMyAdmin showing the stored record in table job_vacancies

6.11 Step 11: Logout

Clicking the Log out button in the header bar ends the session. The registration form is hidden, the success message is cleared, the login boxes are emptied and the user is returned to the login screen. To register another vacancy, the user has to log in again.

CHAPTER 7
TESTING

7.1 Testing Approach

The portal was tested manually through the browser, following the same sequence of steps as a real user. Each screen was checked for correct display, correct messages and correct storage of data. Both valid and invalid inputs were tried so that the positive and the negative behaviour of every validation rule could be observed.

- Unit testing: each field and its rule were checked individually.
- Integration testing: the flow from login to the form, to the database and to logout was checked as a whole.
- Validation testing: wrong lengths, wrong formats, out-of-range numbers and past dates were entered deliberately.
- Interface testing: the layout was checked at different window widths.

7.2 Test Cases

Table 7.1: Test cases and results

No.	Test case	Expected result	Result
1	Start Apache and MySQL from XAMPP	Both modules turn green and show running status	Pass
2	Open the portal at localhost:8080	Login screen is displayed	Pass
3	Login with a valid register number and DOB	Registration form opens	Pass
4	Login with an empty username	Red box and message; form does not open	Pass
5	Login with DOB in a wrong format	Message asks for DD-MM-YYYY	Pass
6	Click Show beside the password	Password is revealed; button changes to Hide	Pass
7	Check Vacancy ID and Posted on	Values are filled automatically and read-only	Pass
8	Register with required fields empty	Empty fields are highlighted; no registration	Pass
9	Enter an invalid e-mail address	E-mail message is shown	Pass
10	Contact number with 9 digits or starting with 5	Rejected with the contact number message	Pass
11	PIN code with 5 digits or starting with 0	Rejected with the PIN code message	Pass
12	Invalid GSTIN; type letters in lower case	Rejected; letters become capitals	Pass
13	Website without http:// or https://	Rejected with the URL message	Pass
14	Number of openings 0 or 501	Rejected with the range message	Pass

No.	Test case	Expected result	Result
15	Maximum salary lower than minimum salary	Rejected with the salary message	Pass
16	Last date to apply in the past	Past dates cannot be selected	Pass
17	Job description below 50 characters	Rejected with the length message	Pass
18	Register without the confirmation tick	Form is not submitted	Pass
19	Fill the form correctly and click Register vacancy	“Vacancy registered successfully”; row stored in job_vacancies	Pass
20	Click Clear form	Boxes are emptied; new Vacancy ID generated	Pass
21	Click Log out	Login screen is shown again	Pass
22	Open submit_job.php directly in the browser	User is sent back to the form page; no row is added	Pass

7.3 Test Observations

The main workflow from login to logout works as designed, and the data submitted through the form is stored correctly in the MySQL database. The validation rules stop wrongly formatted entries before they reach the database. The coloured borders and the messages below the fields make it easy for the user to find and correct a mistake. The stored record can be retrieved instantly, as the phpMyAdmin screen in Figure 7.1 shows.

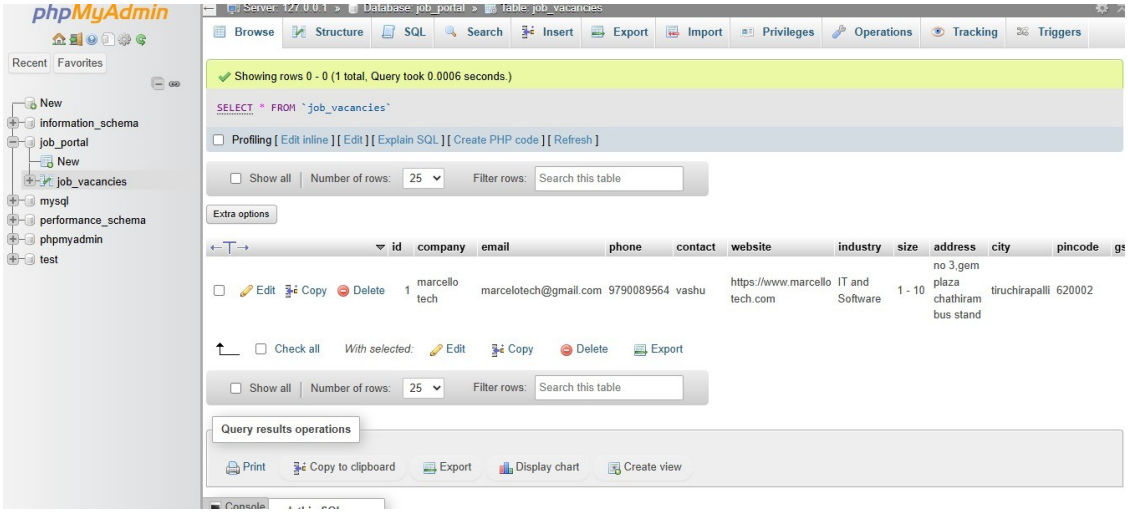


Figure 7.1: Table job_vacancies in phpMyAdmin after registering a vacancy

Observed limitations of the present version are that the login checks only the format of the register number and the date of birth, and that every submission is stored as a new row without a check for duplicate companies. These points are taken up in the future enhancements.

7.4 Sample Test Data

Table 7.2 lists examples of the values that were typed to check the rules. Valid values were accepted and invalid values were rejected with the messages given in Table 5.7.

Table 7.2: Sample valid and invalid test data

Field	Valid value	Invalid value
Username	25UCS552	abc (fewer than 4 characters)
Password (DOB)	15-08-2005	2005-08-15 or 15/08/2005
Official email	hr@company.com	hr@company or hr.company.com
Contact number	9876543210	5876543210 or 98765
Website	https://www.company.com	www.company.com
PIN code	620002	020002 or 62002
GSTIN	33ABCDE1234F1Z5	1234ABCDE5678
Number of openings	1	0 or 501
Salary (min / max)	300000 / 600000	600000 / 300000
Job description	A role description of 50 or more characters	Short text such as “DBA needed”

CHAPTER 8
CONCLUSION AND FUTURE ENHANCEMENT

8.1 Conclusion

The Job Vacancy Registration Portal successfully replaces the informal, e-mail and paper-based way of announcing vacancies with an easy-to-use online process. Using HTML, CSS and JavaScript for the front end, the XAMPP Apache server for hosting and a MySQL database managed with SQL, the system provides a branded login, a complete three-section vacancy form, automatic Vacancy ID and date, strong field validation, a success message, and permanent storage of every vacancy.

The project helped the team to understand web page design, form validation, client-server communication and database handling, and shows how a routine recruitment task can be improved through simple, low-cost technology.

The project also supports the Sustainable Development Goals certified in this report. It supports SDG 8 (Decent Work and Economic Growth) by making it easier for employers to announce jobs, SDG 9 (Industry, Innovation and Infrastructure) by using digital infrastructure for a routine process, and SDG 12 (Responsible Consumption and Production) by reducing the use of paper.

8.2 Limitations of the Present System

- The login checks only the format of the register number and the date of birth and does not compare them with stored records.
- The portal covers only the employer side; candidates cannot search or apply for jobs yet.
- All details of a vacancy are kept in a single table, so company details are repeated if the same company posts many jobs.
- There is no option to edit or delete a vacancy after it has been registered.
- The validation rules run in the browser; submit_job.php cleans the data and the database marks the main columns as required, but the server does not repeat the format checks.
- The database connection uses the default XAMPP user root without a password, which is suitable only for a local computer.

8.3 Future Enhancements

- An admin dashboard for the recruitment team to view, search, approve or reject vacancies.
- Server-side login that checks the register number and date of birth against stored records, with encrypted passwords and OTP verification through e-mail or SMS.
- A candidate module in which job seekers can search vacancies by city, industry, salary or skills, and apply online with a resume.

- Repeating the validation rules in submit_job.php and connecting to the database with a dedicated user that has a password.
- Splitting the data into separate Company, Vacancy, Hiring Process and Benefit tables, as shown in the ER diagram, with a check for duplicate companies.
- E-mail or SMS notification to the company after registration, and automatic closing of a vacancy after its last date.
- Reports and charts showing vacancies by industry, city, job type and salary range.
- Edit and delete options for the vacancies already registered.
- Support for mobile devices through a fully responsive design or a mobile app.

CHAPTER 9
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- phpMyAdmin documentation. https://www.phpmyadmin.net
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